



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Transactions

18

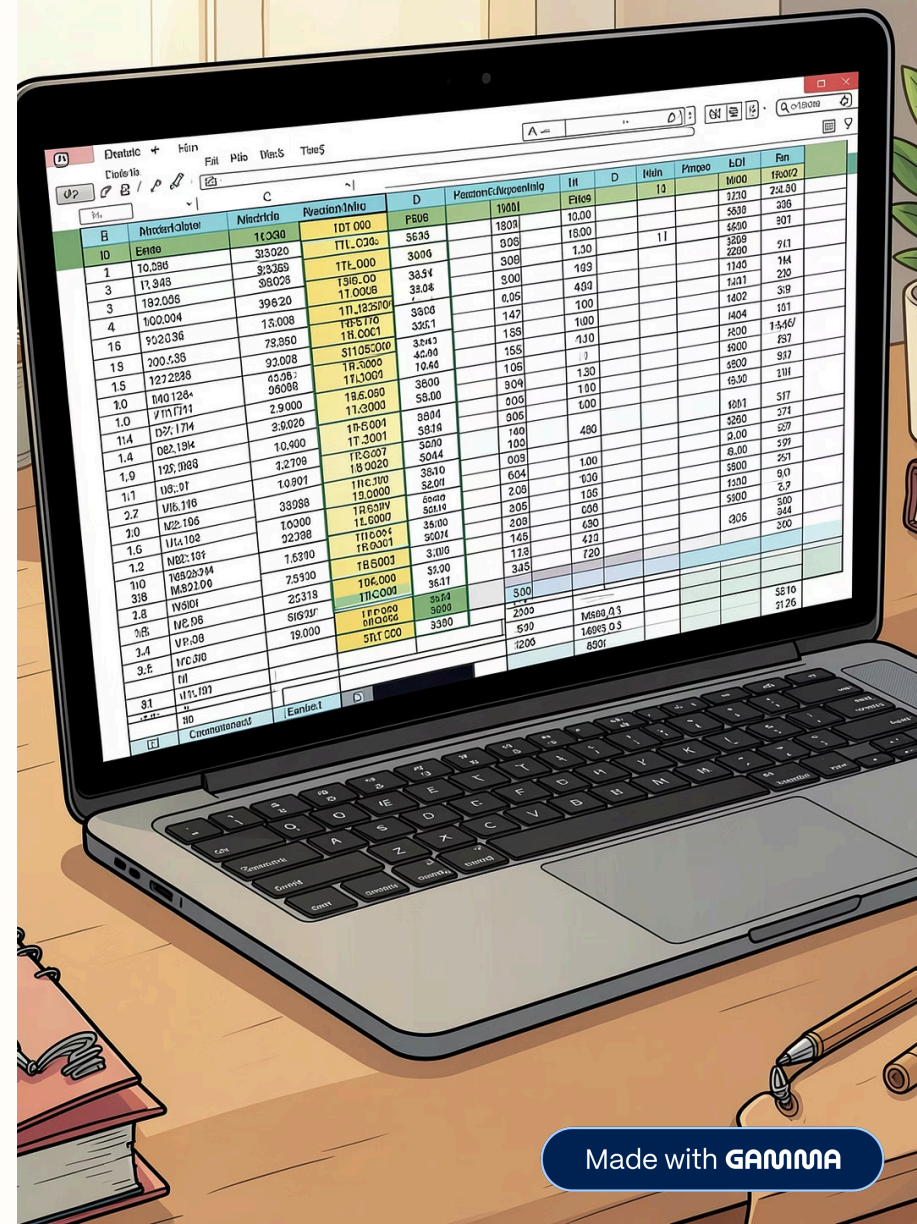
Features

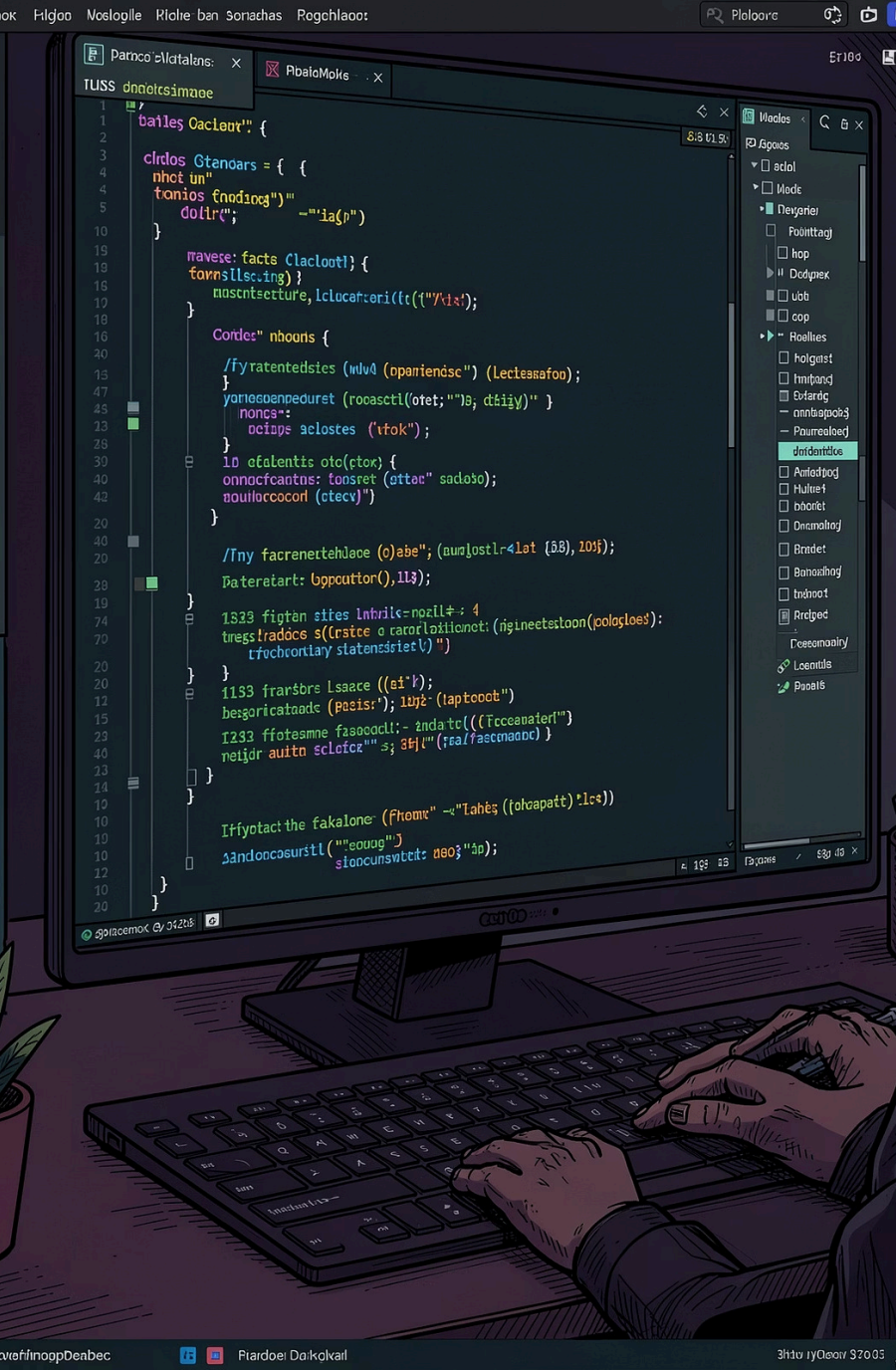
37

Missing Values

Review Rating column only

Demographics, purchase details, and shopping behavior — loaded into PostgreSQL for analysis.





Python Data Preparation

Clean

Imputed missing Review Ratings using median by category

Standardize

Renamed columns to snake_case; dropped redundant promo_code_used

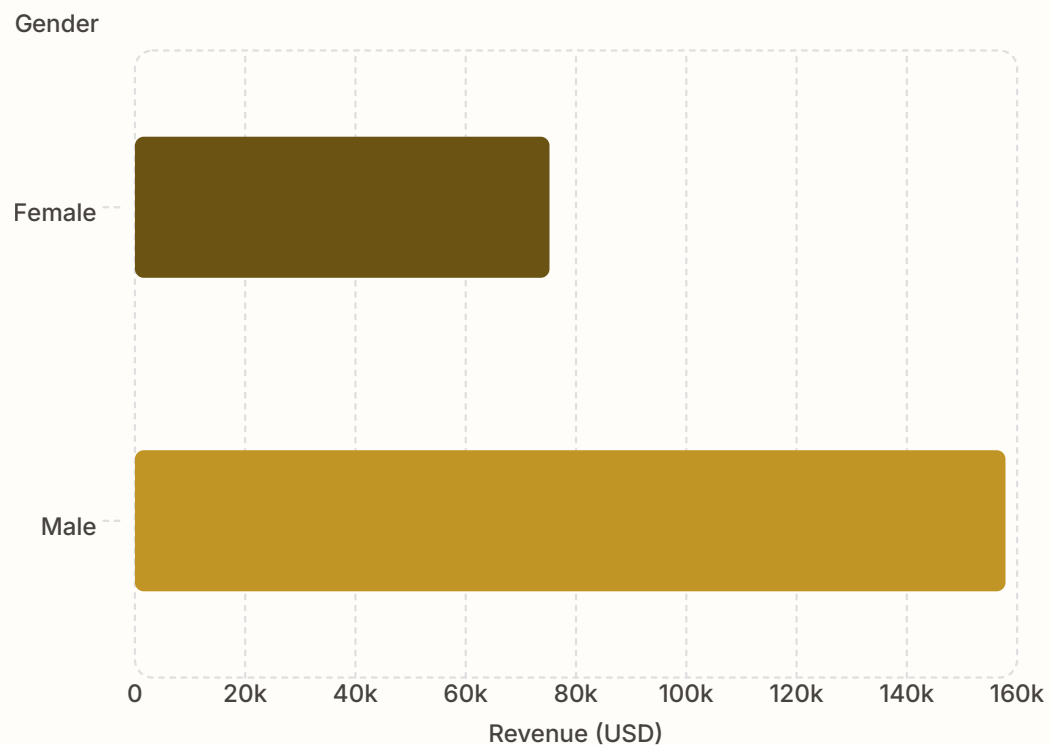
Engineer

Created age_group bins and purchase_frequency_days

Load

Connected to PostgreSQL and loaded cleaned DataFrame

Revenue by Gender



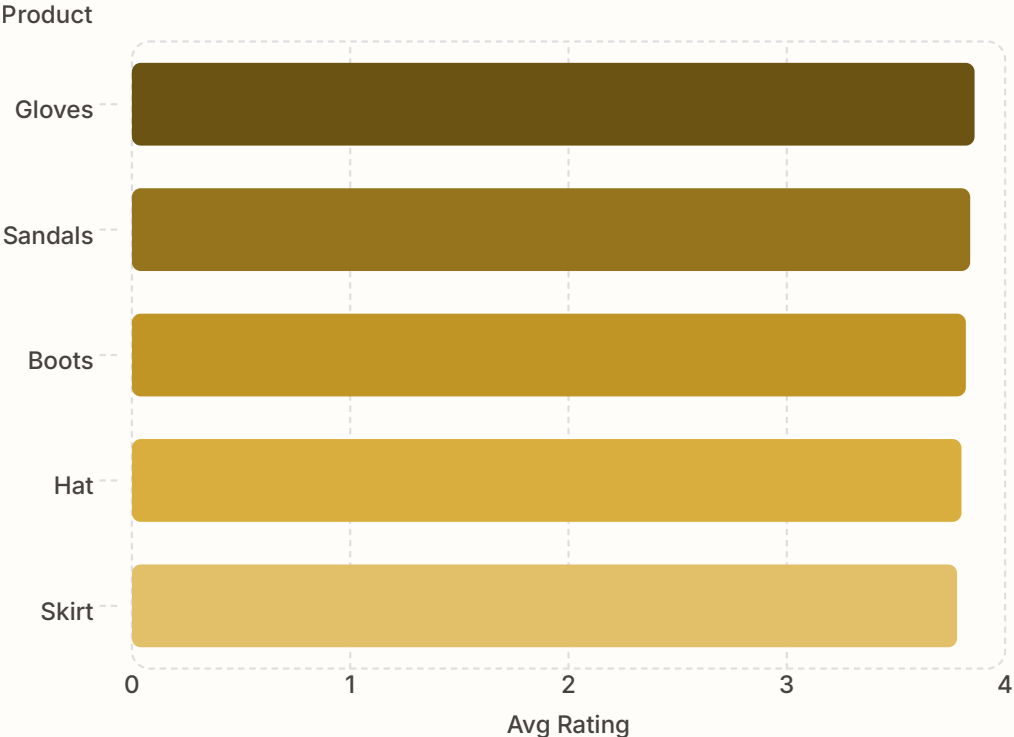
Key Finding

Male customers generated **2.1x more revenue** than female customers — \$157,890 vs \$75,191.

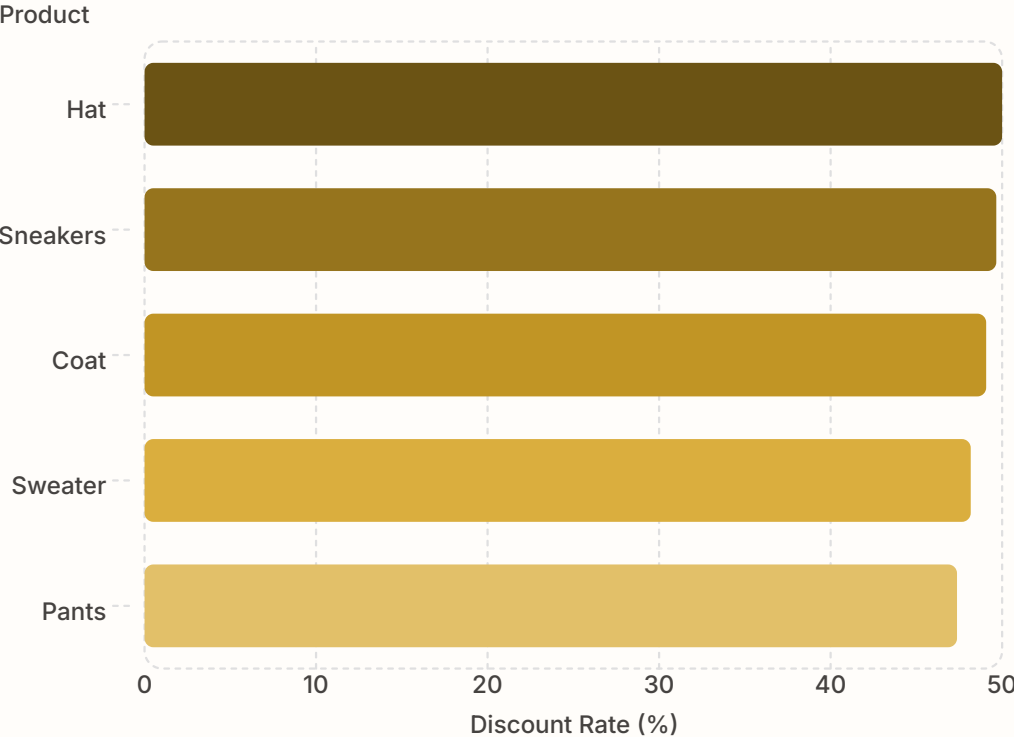
i Male shoppers: 2,652 customers vs 1,248 female.

Top Products & Discount Behavior

Top 5 Products by Rating



Most Discount-Dependent Products

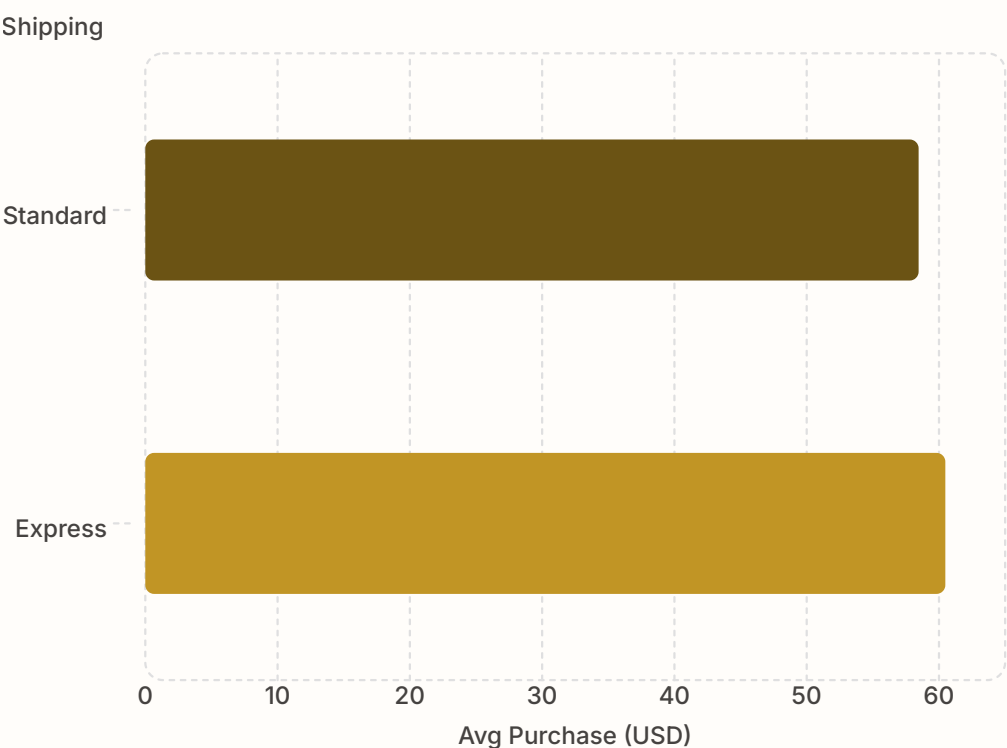


Subscription & Shipping Insights

Revenue: Subscribers vs Non-Subscribers



Avg Purchase by Shipping Type



Non-subscribers drive **73%** of total revenue. Express shipping averages **\$2 more** per order.



Customer Segmentation

3,116

Loyal Customers

80% of base

701

Returning

18% of base

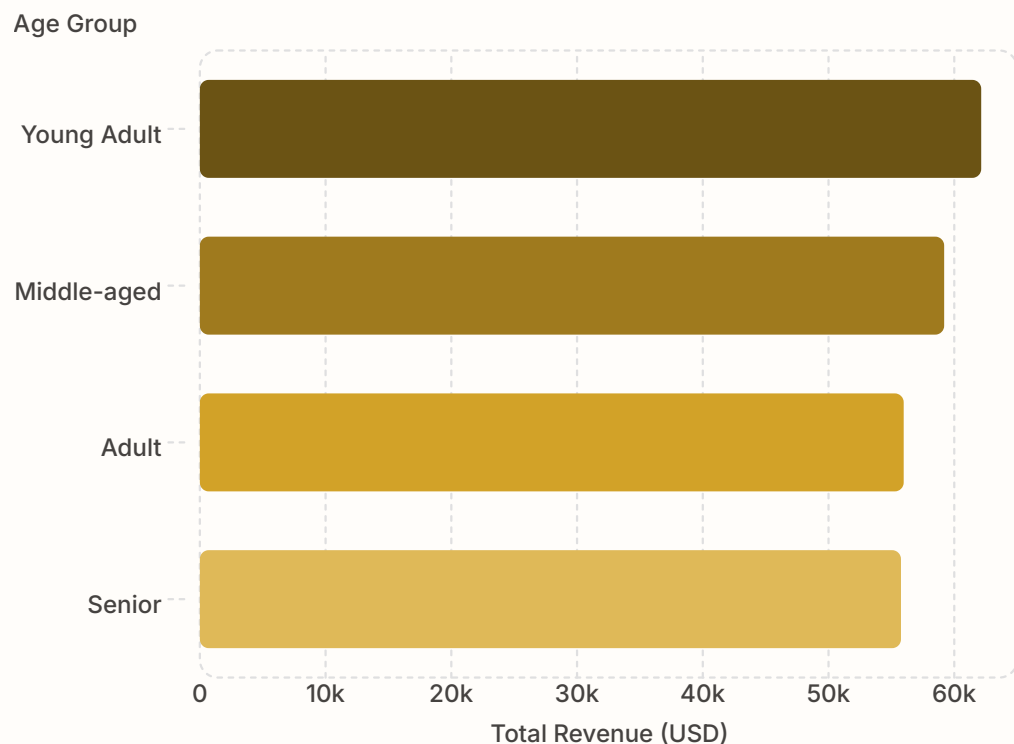
83

New Customers

2% of base

Of 958 repeat buyers (>5 purchases), most remain **non-subscribers** — a key conversion opportunity.

Revenue by Age Group



Key Insight

Young Adults lead revenue contribution. All four age groups are relatively balanced — **no single group dominates**.

Customer age range: **18–70**, mean age 44.

Power BI Dashboard



3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Interactive filters: Subscription Status, Gender, Category, and Shipping Type. Visuals cover revenue, sales, and age group breakdowns.

Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits

→ Loyalty Programs

Reward repeat buyers to grow the Loyal segment

→ Review Discount Policy

Balance sales lift with margin control

→ Product Positioning

Highlight top-rated, best-selling products

→ Targeted Marketing

Focus on high-revenue age groups & express-shipping users